

Block-encoding non-power-of-two matrices

$K = 8$ blocks, phase bitsize $b = 32$.

M (logical row count, possibly non-power-of-2) in $[17, 26, 33, 50, 65, 100, 129, 180, 200, 250]$.

$N = \text{next power of } 2 \geq M$.

SVD-with-padding (this work):

`SVDBlockEncodingInterferometer(n_blocks=K, n_rows=N, n_layers=N, n_rows_logical=M).`

Adds `LessThanConstant(matrix_bitsize, M)` comparators (compute+uncompute, both input and output sides) that OR the $i \geq M$ flag into the block-encoding ancilla. With $\theta_{k,i} = \pi/2$ for $i \geq M$, the block encoding represents the zero-padded $N \times N$ matrix A_{pad} (top-left $M \times M$ is A , else 0).

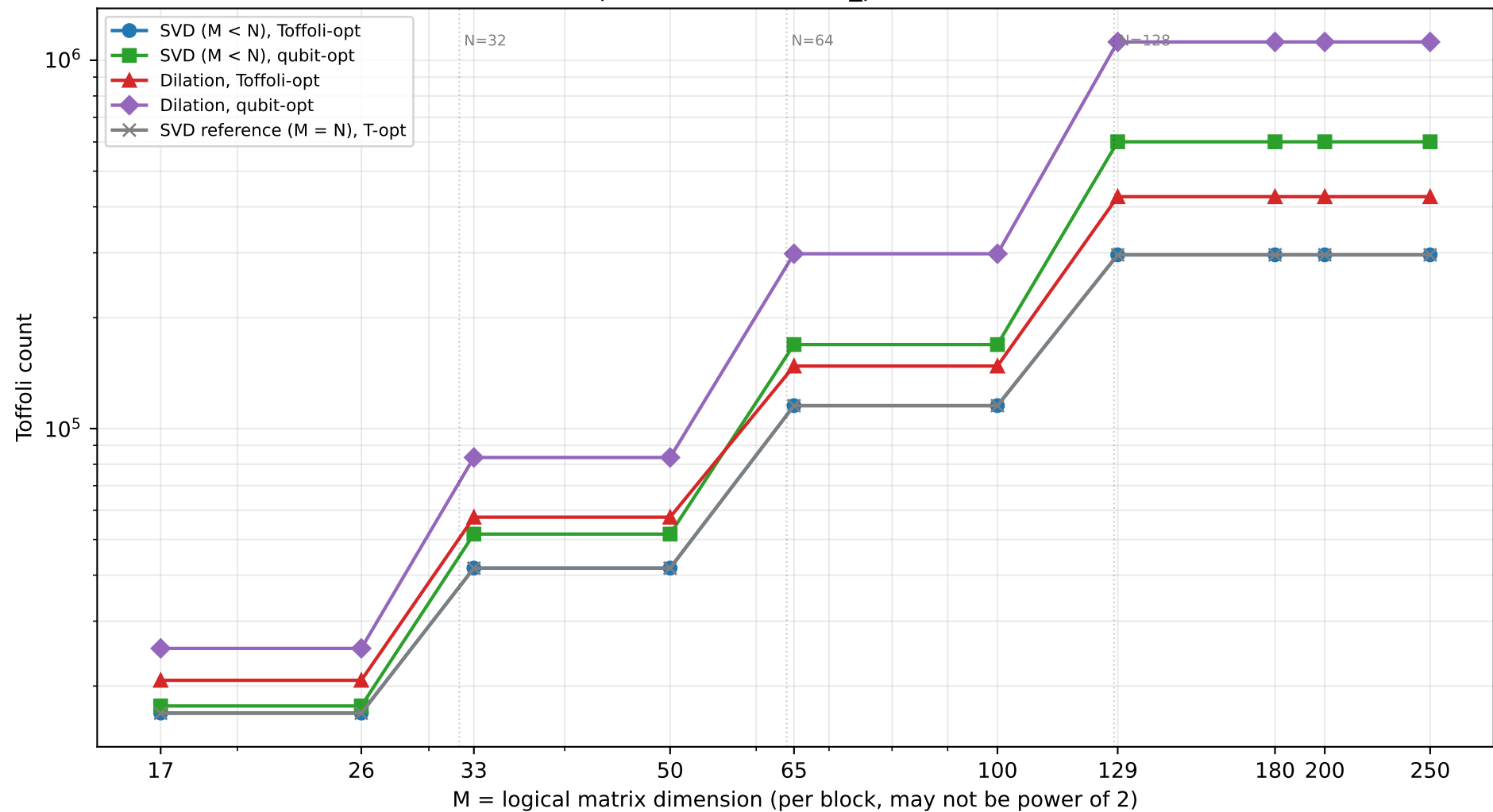
Dilation reference:

Pad A to $N \times N$, Stinespring-dilate to $2N \times 2N$ unitary W , and synthesize $\sum_k |k\rangle\langle k| \otimes W_k$ via `BlockUnitaryInterferometerSynthesisQROAM(K, 2N)`.

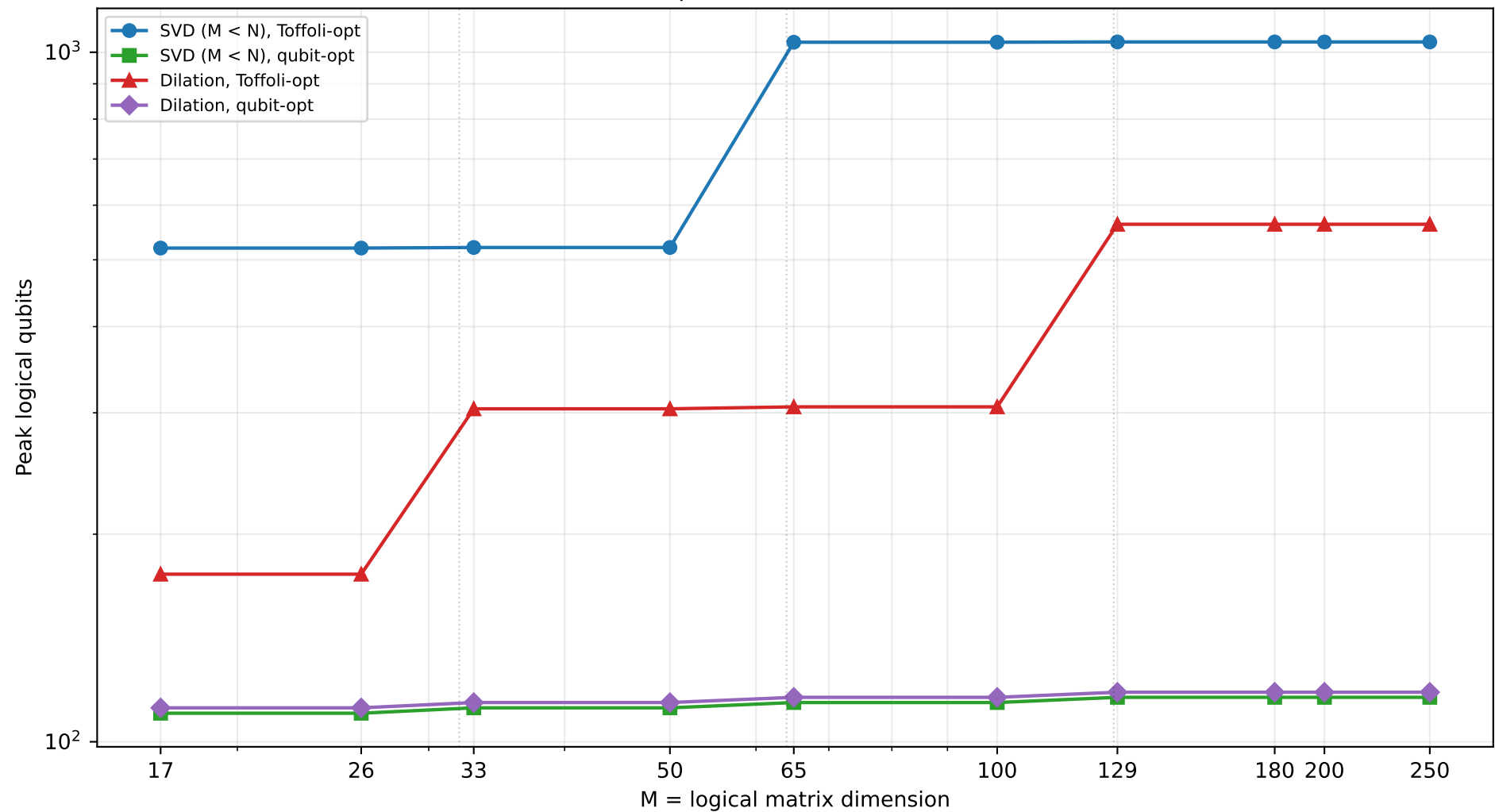
SVD nopad reference: same SVD bloq but with `n_rows_logical=None` (i.e. encoding the synthesized $N \times N$ matrix directly, no comparator projection).

Comparator overhead is two-sided: `4 x LessThanConstant(matrix_bitsize, M)` + `2 x CNOT` (free). Each `LessThanConstant` has Toffoli $\sim \text{matrix_bitsize}$, so the total padding overhead is roughly $4 * \log_2(N)$ Toffolis, which is dominated by the interferometer / QROAM cost (typically $\ll 1\%$ of the total).

Toffoli vs M, padded to $N = \text{next_pow2}(M)$ ($K=8$, $b=32$)



Peak qubits vs M (K=8, b=32)



Resource counts vs M (K=8, b=32)

M	N=next_pow2(M)	SVD-pad T-opt	SVD-pad T-opt	SVD-pad Q-opt	SVD-pad Q-opt	SVD nopad ref T	Dil T-opt T	Dil T-opt Qpadding T	overhead
17	32	16,894	520	17,661	110	16,874	20,730	175	+0.12%
26	32	16,894	520	17,661	110	16,874	20,730	175	+0.12%
33	64	41,810	521	51,713	112	41,786	57,468	304	+0.06%
50	64	41,810	521	51,713	112	41,786	57,468	304	+0.06%
65	128	115,434	1034	169,093	114	115,406	147,836	306	+0.02%
100	128	115,434	1034	169,093	114	115,406	147,836	306	+0.02%
129	256	296,334	1035	600,713	116	296,302	426,240	563	+0.01%
180	256	296,334	1035	600,713	116	296,302	426,240	563	+0.01%
200	256	296,334	1035	600,713	116	296,302	426,240	563	+0.01%
250	256	296,334	1035	600,713	116	296,302	426,240	563	+0.01%